AICE3001/AICE6002: Mathematics of Machine Learning

Subsidiary Notes — Lecture 13: Convexity

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1 Keywords

- Convex sets, convex functions, Jensen's inequality

2 Main Points

2.1 Convex Sets

- We are familiar geometrically with convex regions
- To define convexity we need to define an intermediate point $\boldsymbol{z} = a\,\boldsymbol{x} + (1-a)\,\boldsymbol{y}$
  - we require $a \in [0,1]$ (i.e. a is in the interval between 0 and 1) for $\boldsymbol{z}$ to be between $\boldsymbol{x}$ and $\boldsymbol{y}$
  - to define convexity we only need to have addition and scalar multiplication
- We can then define convexity in a very general way: a set $\mathcal{S}$ is convex if for every pair of points $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{S}$ and for every possible $a \in [0,1]$ then $\boldsymbol{z} = a\,\boldsymbol{x} + (1-a)\,\boldsymbol{y} \in \mathcal{S}$
- We can apply convexity to sets of complicated objects
- For example the set of positive semi-definite matrices forms a convex set
  - This follows from the fact the sum of two positive semi-definite matrices is also positive semi-definite and if we multiply a positive semi-definite matrix by a non-negative number then the matrix is still positive semi-definite

2.2 Convex Functions

- We can define a function, $f(\boldsymbol{x})$, to be a *convex function* if for all pairs of points in the domain of the function and all $a \in [0,1]$

$$f(a\,\boldsymbol{x} + (1-a)\,\boldsymbol{y}) \leq a\,f(\boldsymbol{x}) + (1-a)\,f(\boldsymbol{y})$$

- This means that the function sits on or below the straight chord joining the function values at any two points of its domain
- The epigraph of a function is the area that lies on or above the function
  - The epigraph of a convex function forms a convex region
  - If the epigraph of a function forms a convex region then the function is convex
- We can define *convex-down* or *concave* functions by reversing the inequality

$$f(a\,\boldsymbol{x} + (1-a)\,\boldsymbol{y}) \geq a\,f(\boldsymbol{x}) + (1-a)\,f(\boldsymbol{y})$$

- for clarity I will sometimes refer to "convex" functions as *convex-up* functions
- convex-down functions have similar (but opposite) properties to convex up functions

- A function where for every pair of distinct points $\boldsymbol{x} \neq \boldsymbol{y}$ and for any a such that $0 < a < 1$ (i.e. a lies strictly between 0 and 1)

$$f(a\,\boldsymbol{x} + (1-a)\,\boldsymbol{y}) < a\,f(\boldsymbol{x}) + (1-a)\,f(\boldsymbol{y})$$

  is said to be *strictly convex*

- Linear functions $f(\boldsymbol{x}) = \boldsymbol{m}^{\mathsf{T}}\boldsymbol{x} + c$ are both convex-up and convex-down functions
  - For a function to be a strictly convex function it cannot have a linear section

- Convex functions lie on or above their tangent plane (where they are differentiable)

$$f(\boldsymbol{x}) \geq f(\boldsymbol{x}_0) + (\boldsymbol{x} - \boldsymbol{x}_0)^{\mathsf{T}}\boldsymbol{\nabla} f(\boldsymbol{x}_0)$$

  - The tangent plane at $\boldsymbol{x}_0$ is the linear function that matches both the value $f(\boldsymbol{x}_0)$ and the gradient $\boldsymbol{\nabla} f(\boldsymbol{x}_0)$ there. (The gradient is orthogonal to the *contour* through $\boldsymbol{x}_0$, not to the tangent plane)
  - Where f is not differentiable (e.g. $|x|$ at $x = 0$) there is no unique tangent, but a convex function still lies above some line through each point (a *supporting* line, whose slope is called a *subgradient*). This is enough for everything below, including Jensen's inequality

- Why the second derivative is non-negative: the tangent lines at x and $x + \epsilon$ (with $\epsilon > 0$) give

$$\begin{aligned} f(x+\epsilon) &\geq f(x) + \epsilon\,f'(x) \\ f(x) &\geq f(x+\epsilon) - \epsilon\,f'(x+\epsilon) \end{aligned}$$

  Adding these, $0 \geq \epsilon\left(f'(x) - f'(x+\epsilon)\right)$, so $f'(x+\epsilon) \geq f'(x)$: the slope never decreases, and hence $f''(x) \geq 0$

- A necessary and sufficient condition for a twice-differentiable function to be convex is that its second derivative is non-negative or for multi-dimensional functions the Hessian is positive semi-definite
  - If the second derivative is positive (i.e. always greater than 0) or the Hessian is positive definite then the function is strictly convex
  - This is sufficient but not necessary: x^4 is strictly convex although its second derivative vanishes at $x = 0$

- Examples
  - Convex-up Functions
    - $f(x) = x^2$ is strictly convex since $f''(x) = 2 > 0$
    - $f(x) = x^{-2}$ (for $x > 0$) is strictly convex since $f''(x) = 6\,x^{-4} > 0$
    - $f(x) = x^4$ is convex since $f''(x) = 12\,x^2 \geq 0$ (in fact strictly convex, see above)
    - $f(x) = \mathrm{e}^{c\,x}$ is strictly convex for $c \neq 0$ as $f''(x) = c^2\,\mathrm{e}^{c\,x} > 0$
    - $f(\boldsymbol{x}) = \|\boldsymbol{x}\|^2$ is strictly convex since $\mathbf{H} = 2\,\mathbf{I} \succ 0$
    - $f(x) = |x|$ is convex since for $a \in [0,1]$ we have $|a\,x + (1-a)\,y| \leq a\,|x| + (1-a)\,|y|$ (the triangular inequality), with equality when $x\,y \geq 0$, so it is not strictly convex
    - $f(\boldsymbol{x}) = \|\boldsymbol{x}\|_1 = \sum_i |x_i|$ is convex, as a sum of convex functions

- $f(x, y) = x^2/y$ (for $y > 0$) is convex: its Hessian

$$\mathbf{H} = \frac{2}{y^3}\begin{pmatrix} y^2 & -x\,y \\ -x\,y & x^2 \end{pmatrix}$$

  has eigenvalues 0 and $2(x^2 + y^2)/y^3$, both non-negative

- Convex-down Functions
  1. $f(x) = -x^2$ is strictly convex-down since $f''(x) = -2 < 0$
  2. $f(x) = \sqrt{x}$ (for $x > 0$) is strictly convex-down since $f''(x) = -x^{-3/2}/4 < 0$
  3. $f(x) = \log(x)$ is strictly convex-down since $f''(x) = -1/x^2 < 0$

- A function $f(\boldsymbol{x})$ that is constrained to a convex domain ($\boldsymbol{x} \in \mathcal{S}$, where $\mathcal{S}$ is a convex set) is convex in that domain if for all pairs $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{S}$ and all $a \in [0, 1]$ we have

$$f(a\,\boldsymbol{x} + (1 - a)\,\boldsymbol{y}) \leq a\,f(\boldsymbol{x}) + (1 - a)\,f(\boldsymbol{y})$$

  - This is just a more precise definition of a convex function
  - Note that by limiting the domain of a function some non-convex functions may be convex over that domain
    - e.g. $\cos(x)$ is convex in the interval $[\pi/2, 3\pi/2]$, where $-\cos(x) \geq 0$, and $\sin(x)$ is convex-down in $[0, \pi]$
  - A convex function constrained to lie in a convex set will still be convex
  - Any combination of linear constraints will form a convex set
  - Non-linear constraints may or may not: with $\mathbf{M} \succeq 0$, $\boldsymbol{x}^\mathsf{T}\mathbf{M}\boldsymbol{x} \leq 1$ is a convex set (an ellipsoid) but $\boldsymbol{x}^\mathsf{T}\mathbf{M}\boldsymbol{x} \geq 1$ is not
  - Therefore convex functions restricted to satisfy linear constraints will be convex

- The minima of a convex function (if it has any) form a convex set
  - There can be no local minima that are not global: on the line from such a point to a lower point the function lies on or below the chord, so there would be lower points arbitrarily close by
  - A strictly convex function has at most one minimum: if $\boldsymbol{x} \neq \boldsymbol{y}$ were both minima, strict convexity would make $f((\boldsymbol{x} + \boldsymbol{y})/2)$ smaller than $f(\boldsymbol{x})$. It need not have one at all (e^x has none)

- A sum of convex functions with non-negative weights is convex
  - Adding a strictly convex function to a convex one gives a strictly convex function (add the two defining inequalities; one is strict)

- **Linear regression**
  - The loss function of linear regression is convex

$$L(\boldsymbol{w}) = \|\mathbf{X}\boldsymbol{w} - \boldsymbol{y}\|^2$$

    - The Hessian is $2\,\mathbf{X}^\mathsf{T}\mathbf{X}$ which is positive semi-definite which is a sufficient condition for $L(\boldsymbol{w})$ to be convex
  - Both the L_2 regulariser and the L_1 regulariser are convex
  - The L_2 regulariser is strictly convex so adding it (ridge regression, with $\eta > 0$) gives a unique solution. So does the elastic net (L_1 plus L_2), while the L_1 regulariser alone (lasso) gives a convex loss whose minimum need not be unique
  - Many machine learning algorithms are chosen because they involve minimising a convex function leading to a unique minimum

2.3 Jensen's Inequality

- For any convex-up function, if $\boldsymbol{x}$ is a random variable then

$$\mathbb{E}[f(\boldsymbol{x})] \geq f(\mathbb{E}[\boldsymbol{x}])$$

  - $\mathbb{E}[\cdots]$ denotes the expectation

- For any convex-down function

$$\mathbb{E}[f(\boldsymbol{x})] \leq f(\mathbb{E}[\boldsymbol{x}])$$

- These are known as *Jensen's Inequality*

- **Proof**

  - We can prove this (for differentiable f; a supporting plane does the same job otherwise) starting from the fact that $f(\boldsymbol{x})$ lies above the tangent plane at any point

$$f(\boldsymbol{x}) \geq f(\boldsymbol{x}^*) + (\boldsymbol{x} - \boldsymbol{x}^*)^\mathsf{T} \boldsymbol{\nabla} f(\boldsymbol{x}^*)$$

  - This has to be true at the point $\boldsymbol{x}^* = \mathbb{E}[\boldsymbol{x}]$

$$f(\boldsymbol{x}) \geq f(\mathbb{E}[\boldsymbol{x}]) + (\boldsymbol{x} - \mathbb{E}[\boldsymbol{x}])^\mathsf{T} \boldsymbol{\nabla} f(\mathbb{E}[\boldsymbol{x}])$$

  - Taking expectations of both sides of the equation

$$\begin{aligned} \mathbb{E}[f(\boldsymbol{x})] &\geq f(\mathbb{E}[\boldsymbol{x}]) + (\mathbb{E}[\boldsymbol{x}] - \mathbb{E}[\boldsymbol{x}])^\mathsf{T} \boldsymbol{\nabla} f(\mathbb{E}[\boldsymbol{x}]) \\ &= f(\mathbb{E}[\boldsymbol{x}]) \qquad\qquad \square \end{aligned}$$

- Using Jensen's Inequality

  - Consider the strictly convex function $f(x) = x^2$ by Jensen's inequality

$$\mathbb{E}\left[x^2\right] \geq \mathbb{E}[x]^2$$

    - or $\mathbb{E}\left[x^2\right] - \mathbb{E}[x]^2 \geq 0$
      - the left-hand side is the variance so we see variances are non-negative
      - because $f(x) = x^2$ is strictly convex we only get equality where $\boldsymbol{x}$ doesn't vary at all

  - Consider the Kullback-Leibler (KL) divergence defined for discrete probability distributions defined over the same range as

$$\mathrm{KL}(f \| g) = -\sum_i f_i \log\left(\frac{g_i}{f_i}\right)$$

    - This is often used to measure how different distribution are from each other
    - Note if $g_i = f_i$ then $\mathrm{KL}(f \| g) = 0$ since $\log(1) = 0$
    - Now we can use Jensen's inequality to show that $\mathrm{KL}(f \| g) \geq 0$

$$\begin{aligned} \mathrm{KL}(f \| g) &= -\sum_i f_i \log\left(\frac{g_i}{f_i}\right) = -\mathbb{E}_f\left[\log\left(\frac{g_i}{f_i}\right)\right] \\ &\geq -\log\left(\mathbb{E}_f\left[\frac{g_i}{f_i}\right]\right) \\ &= -\log\left(\sum_i f_i \frac{g_i}{f_i}\right) = -\log\left(\sum_i g_i\right) = -\log(1) = 0 \end{aligned}$$

        * Here we are assuming we have random variable that take values $X_i = g_i/f_i$ that occur with probability f_i
        * The KL-divergence is therefore equal to $\mathbb{E}[-\log(X_i)]$
        * Since $-\log(x)$ is convex up we have by Jensen's inequality that the KL-divergence is greater than or equal to $-\log(\mathbb{E}[X_i]) = -\log(\sum_i f_i\, X_i)$
        * But $X_i = g_i/f_i$ so the KL-divergence is greater than or equal to $-\log(\sum_i g_i)$
        * But g_i is a probability so $\sum_i g_i = 1$ giving us our result
        * If some $f_i = 0$ we take $0\log(0) = 0$, and X_i is only defined where $f_i > 0$, so every sum runs over those i. Then $\sum_{i:f_i>0} g_i \leq 1$, so $-\log$ of it is still ≥ 0 and the result holds
    - This is known as the Gibbs' inequality after the mathematical physicist, J. Willard Gibbs, (founder of modern statistical mechanics) who first proved it
    - We often use KL-divergences when we want to choose the parameters of one probability distribution so that it approximates a second probability distribution

3 Exercises

3.1 Positive quadrant

- Prove that the set of vectors with non-negative elements form a convex set

3.2 Inverse of Convex Functions

1. Use the chain rule to compute the second derivative of $f(g(x))$
2. If $g(x) = f^{-1}(x)$ show that the second derivative of $f(g(x))$ vanishes
3. Use these results to derive an identity for the second derivative of $f^{-1}(x)$
4. Derive a condition for $f^{-1}(x)$ to be a convex-down function given that $f(x)$ is convex-up
5. Use this to show
   (a) $\sqrt{x}$ is a convex-down function
   (b) $\log(x)$ is a convex-down function

3.3 Cumulant Generating Function

- Here is something a bit harder (which you don't need to know)
- The cumulant generating function of a probability distribution $p(x)$ is defined as

$$G(\lambda) = \log\Big(\mathbb{E}\left[\mathrm{e}^{\lambda\, x}\right]\Big)$$

    - the expectation is over the random variable x drawn from $p(x)$

- It is called the cumulant generating function because if we take the nth derivative and set λ to zero we obtain the nth cumulant (i.e. $\kappa_n = G^{(n)}(0)$)
- The first cumulant is the mean, the second the variance while the third and fourth are proportional to the skewness and excess kurtosis
- Cumulant generating functions appear a lot when you work with probabilities, but go beyond this course
- Nevertheless let's show they are convex

1. Find the second derivative
2. Show that if $p(x)$ is a probability distribution then $q(x) = p(x)\,\mathrm{e}^{\lambda\,x}/\mathbb{E}\left[\mathrm{e}^{\lambda\,x}\right]$ is also a probability distribution
3. Hence show that the cumulant generating function is convex

- See answers

4 Answers

4.1 Positive quadrant

- Let $\mathcal{P}$ be the set of vectors with non-negative elements
- If $\boldsymbol{x} \in \mathcal{P}$ then if $c \geq 0$ we have $\boldsymbol{v} = c\,\boldsymbol{x} \in \mathcal{P}$ since each element of $\boldsymbol{v}$ will be non-negative (i.e. $v_i = c\,x_i \geq 0$)
- Also for any two vectors $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{P}$ clearly $\boldsymbol{w} = \boldsymbol{x} + \boldsymbol{y} \in \mathcal{P}$ since $w_i = x_i + y_i$
- Thus for any two vectors $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{P}$ and any $a \in [0, 1]$ the vector $\boldsymbol{z} = a\,\boldsymbol{x} + (1-a)\,\boldsymbol{y}$ will be in $\mathcal{P}$

4.2 Inverse of Convex Functions

1. Taking derivatives

$$\frac{\mathrm{d}^2 f(g(x))}{\mathrm{d}x^2} = \frac{\mathrm{d}f'(g(x))\,g'(x)}{\mathrm{d}x} = f''(g(x))\,(g'(x))^2 + f'(g(x))\,g''(x)$$

2. If $g(x) = f^{-1}(x)$ then $f(g(x)) = x$ and the second derivative vanishes
3. Using 1. and 2. we find (writing $f^{-1}(x)$ as $g(x)$)

$$g''(x) = -\frac{f''(g(x))\,(g'(x))^2}{f'(g(x))}$$

4. If $f(x)$ is convex then $f''(y) \geq 0$ for any y (including $y = f^{-1}(x)$) also $(g'(x))^2 \geq 0$ so for the inverse of $f(x)$ to be convex down we require $f'(f^{-1}(x)) > 0$
5. Use this to show
   (a) Let $f(x) = x^2$ for $x \geq 0$, so that $f''(x) = 2 > 0$ and $f'(y) = 2\,y$ which is non-negative if $y \geq 0$, but $f^{-1}(x) = \sqrt{x} > 0$ so $f'(f^{-1}(x)) \geq 0$ and consequently $\sqrt{x}$ is convex-down
   (b) Let $f(x) = \exp(x)$, so that $f''(x) = \exp(x) > 0$. But $f'(y) = \exp(y) > 0$ for all y so $f'(f^{-1}(x)) > 0$ which is sufficient to show $f^{-1}(x) = \log(x)$ is a convex-down function

4.3 Cumulant Generating Function

1. If $G(\lambda) = \log\bigl(\mathbb{E}\left[\mathrm{e}^{\lambda\,x}\right]\bigr)$ then

$$G'(\lambda) = \frac{\mathbb{E}\left[x\,\mathrm{e}^{\lambda\,x}\right]}{\mathbb{E}\left[\mathrm{e}^{\lambda\,x}\right]}$$

and

$$G''(\lambda) = \frac{\mathbb{E}\left[x^2\,\mathrm{e}^{\lambda\,x}\right]}{\mathbb{E}\left[\mathrm{e}^{\lambda\,x}\right]} - \frac{\mathbb{E}\left[x\,\mathrm{e}^{\lambda\,x}\right]^2}{\mathbb{E}\left[\mathrm{e}^{\lambda\,x}\right]^2}$$

2. - Now if $p(x)$ is a probability distribution it will be non-negative for all x and sum or integrate to 1

- But then $q(x) = p(x)\,\mathrm{e}^{\lambda x}/\mathbb{E}\left[\mathrm{e}^{\lambda x}\right]$ will be non-negative as $\mathrm{e}^{\lambda x} > 0$ and $\mathbb{E}\left[\mathrm{e}^{\lambda x}\right] > 0$ (the expectation of positive quantities will be positive)
- But
$$\int q(x)\,\mathrm{d}x = \frac{1}{\mathbb{E}\left[\mathrm{e}^{\lambda x}\right]} \int p(x)\,\mathrm{e}^{\lambda x}\,\mathrm{d}x = \frac{\mathbb{E}\left[\mathrm{e}^{\lambda x}\right]}{\mathbb{E}\left[\mathrm{e}^{\lambda x}\right]} = 1$$
- So $q(x)$ is non-negative and normalised so is a well defined probability distribution

3. Using the result of 1. and 2
$$G''(\lambda) = \frac{\mathbb{E}_p\left[x^2\,\mathrm{e}^{\lambda x}\right]}{\mathbb{E}_p\left[\mathrm{e}^{\lambda x}\right]} - \frac{\mathbb{E}_p\left[x\,\mathrm{e}^{\lambda x}\right]^2}{\mathbb{E}_p\left[\mathrm{e}^{\lambda x}\right]^2} = \mathbb{E}_q\left[x^2\right] - \mathbb{E}_q[x]^2 \geq 0$$
- since variances are non-negative